

Foundations of BLEU score

[BLEU score]

1.0 Content Block

Final formula There is not a single definition of BLEU, but a whole family of them, parametrized by the weighting vector $w := (w_1, w_2, \dots)$. It is a probability distribution on $\{1, 2, 3, \dots\}$, that is, $\sum_{i=1}^{\infty} w_i = 1$, and $\forall i \in \{1, 2, 3, \dots\}, w_i \in [0, 1]$. With a choice of w , the BLEU score is $BLEU_w(\hat{S}; S) := BP(\hat{S}; S) \cdot \exp\left(\sum_{n=1}^{\infty} w_n \ln p_n(\hat{S}; S) - \ln p_n(S)\right)$. In words, it is a weighted geometric mean of all the modified n-gram precisions, multiplied by the brevity penalty. We use the weighted geometric mean, rather than the weighted arithmetic mean, to strongly favor candidate corpuses that are simultaneously good according to multiple n-gram precisions. The most typical choice, the one recommended in the original paper, is $w_1 = w_4 = 1/4$.

2.0 Content Block

Performance BLEU has frequently been reported as correlating well with human judgement,[10][11][12] and remains a benchmark for the assessment of any new evaluation metric. There are however a number of criticisms that have been voiced. It has been noted that, although in principle capable of evaluating translations of any language, BLEU cannot, in its present form, deal with languages lacking word boundaries.[13] Designed to be used for several reference translation, in practice it's used with only the single one. BLEU is infamously dependent on the tokenization technique, and scores achieved with different ones are incomparable (which is often overlooked); in order to improve reproducibility and comparability, SacreBLEU variant was designed. It has been argued that although BLEU has significant advantages, there is no guarantee that an increase in BLEU score is an indicator of improved translation quality.[14]

3.0 Content Block

where m is number of words from the candidate that are found in the reference, and w is the total number of words in the candidate. This is a perfect score, despite the fact that the candidate translation above retains little of the content of either of the references. The modification that BLEU makes is fairly straightforward. For each word in the candidate translation, the algorithm takes its maximum total count, $m_{\max}$, in any of the reference translations. In the example above, the word "the" appears twice in reference 1, and once in reference 2. Thus $m_{\max} = 2$. For the candidate translation, the count m_w of each word is clipped to a maximum of $m_{\max}$ for that word. In this case, "the" has $m_w = 7$ and $m_{\max} = 2$, thus m_w is clipped to 2. These clipped counts m_w are then summed over all distinct words in the candidate. This sum is then divided by the total number of unigrams in the candidate translation. In the above example, the modified unigram precision score would be:

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as the word 'the' and the word 'cat' appear once each in the candidate, and the total number of words is two. The modified bigram precision would be $1/1$ as the bigram, "the cat" appears once in the candidate. It has been pointed out that precision is usually twinned with recall to overcome this problem [7], as the unigram recall of this example would be $3/6$ or $2/7$

2/7}. The problem being that as there are multiple reference translations, a bad translation could easily have an inflated recall, such as a translation which consisted of all the words in each of the references.[8] To produce a score for the whole corpus, the modified precision scores for the segments are combined using the geometric mean multiplied by a brevity penalty to prevent very short candidates from receiving too high a score. Let r be the total length of the reference corpus, and c the total length of the translation corpus. If $c \leq r$, the brevity penalty applies, defined to be $e^{(1-r/c)}$. (In the case of multiple reference sentences, r is taken to be the sum of the lengths of the sentences whose lengths are closest to the lengths of the candidate sentences. However, in the version of the metric used by NIST evaluations prior to 2009, the shortest reference sentence had been used instead.) iBLEU is an interactive version of BLEU that allows a user to visually examine the BLEU scores obtained by the candidate translations. It also allows comparing two different systems in a visual and interactive manner which is useful for system development.[9]

5.0 Content Block

c is the length of the candidate corpus, that is, $c := \sum_{i=1}^M |y^{(i)}|$ where $|y|$ is the length of y .

6.0 Content Block

Modified n-gram precision Define the modified n-gram precision function to be $p_n(\hat{S}; S) := \sum_{i=1}^M \sum_{s \in G_n(y^{(i)})} \min(C(s, y^{(i)}), \max_{y \in S} C(s, y))$

$$p_n(\hat{S}; S) = \frac{\sum_{i=1}^M \sum_{s \in G_n(\hat{y}^{(i)})} \min(C(s, \hat{y}^{(i)}), \max_{y \in S} C(s, y))}{\sum_{i=1}^M \sum_{s \in G_n(\hat{y}^{(i)})} C(s, \hat{y}^{(i)})}$$

The modified n-gram, which looks complicated, is merely a straightforward generalization of the prototypical case: one candidate sentence and one reference sentence. In this case, it is $p_n(\hat{y}; y) = \sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), C(s, y)) / \sum_{s \in G_n(\hat{y})} C(s, \hat{y})$

$$p_n(\hat{y}; y) = \frac{\sum_{s \in G_n(\hat{y})} \min(C(s, \hat{y}), C(s, y))}{\sum_{s \in G_n(\hat{y})} C(s, \hat{y})}$$

To work up to this expression, we start with the most obvious n-gram count summation: $\sum_{s \in G_n(\hat{y})} C(s, y) = \text{number of n-substrings in } \hat{y} \text{ that appear in } y$

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